
In CEM, a World Model Is Also a Proposal Mechanism

Oliver Obst^{1*}

Frieder Stolzenburg^{2,1†}

¹UNSW Sydney

²Harz University of Applied Sciences

Abstract

The cross-entropy method (CEM) uses world-model scores to select action sequences and fit the distribution sampled in its next iteration. A scoring error can therefore change both the present decision and the candidates considered later. We evaluate these two roles separately. Four types of predictive model generate CEM traces, and every model rescores every saved candidate pool. Executing the same candidates in the environment provides a reference elite set and proposal update.

Across twelve independently trained task-seed units on Walker and Cheetah, the pre-specified proposal distance falls from the first to the final CEM iteration in every unit. Proposal widths contract and fitted means separate relative to the remaining search width. Pairwise ranking agreement stays near chance on Walker and declines on Cheetah; elite-set agreement does not improve. This comparison shows greater variation between scorers than between pool sources on Cheetah; Walker has variation in both and in their pairings. We use the original six units to select Random nonlinear for a one-update intervention, without inspecting intervention outcomes. Replacing its first model-ranked update with an environment-ranked update lowers final realised selected-sequence cost in those six units and in six further units held out from the selection.

1 Introduction

A world model used for planning determines which actions receive further consideration. In model-predictive control (MPC), the planner predicts the outcomes of candidate action sequences, selects a sequence, and executes its first action before replanning. The cross-entropy method (CEM) performs this search by repeatedly sampling candidates and fitting a new sampling distribution to those with the lowest predicted cost [Bharadhwaj et al., 2020]. We call that distribution the *proposal* and the retained candidates the *elite set*.

As an illustrative example, consider two candidate pools for the same physical state and cost objective. Pool A contains a sequence with realised cost 1, but the scorer selects one with realised cost 5. This is a selection error: a lower-cost sequence was available. Pool B contains only sequences with realised cost 8 or higher. Its scorer orders every candidate correctly and selects the cost-8 sequence. Earlier CEM updates could have left the planner searching this second region. Correct ordering within that pool still produces a higher-cost choice than the misranked choice from pool A.

A fixed-pool comparison with environment executions can measure ranking errors, elite membership, proposal-refit disagreement, and selection regret among the available candidates. In the example, it can expose the error in pool A and confirm the ordering in pool B. It cannot show which lower-cost sequences another proposal would have sampled. Correct within-pool ordering does not ensure that the pool contains low-cost actions.

*o.obst@unsw.edu.au

†fstolzenburg@hs-harz.de

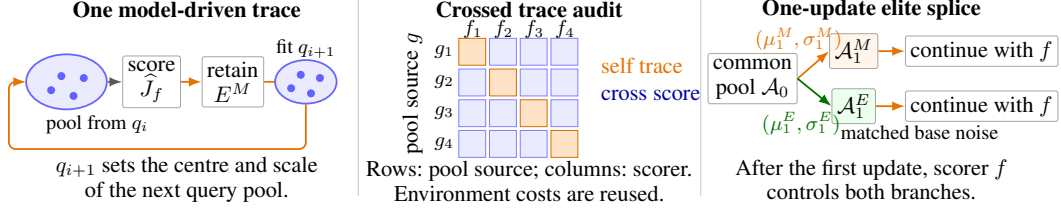


Figure 1: CEM feedback, crossed rescoring, and the one-update splice. Left: a model’s elite set fits the next proposal, changing the source of later candidates. Centre: each row holds an executed pool fixed across scorers; each column holds a scorer fixed across pool sources. The diagonal is the usual model-driven search. Right: the splice replaces only the first elite set, then matches the scorer and base random samples in both branches. The crossed comparison measures disagreement on existing pools; the splice changes which pools are sampled next.

Evaluation also depends on where the candidates came from. Logged rollouts use a data-collection policy; actor-centred ranking tests perturb a reference action sequence. These tests can measure prediction or ordering on their chosen action distributions. CEM changes its own distribution after each elite decision. We call this *planner-induced query shift*: model scores change the optimiser’s future queries. The physical system need not change.

Rank agreement measured along a CEM trace already captures how ordering changes during search [Wang et al., 2026]. Our comparison adds pools produced by other models’ update histories. Applying the same scorer to those pools tests whether its disagreement depends on pool source; applying different scorers to one pool compares their decisions on identical actions. Realised candidate costs separately record which pools contain lower-cost sequences.

Our comparison uses four predictors fitted over frozen Dreamer contexts [Hafner et al., 2020]. A context is the backbone’s internal representation of the observation history. Each predictor replaces the full Dreamer dynamics during planning, so we call it a *surrogate*. It predicts future contexts under a candidate action sequence. A fixed readout maps those contexts to the task variables used in the planning cost. The four surrogates share data, transition form, cost, and CEM settings; their feature maps differ.

Each surrogate first runs its own CEM search. We save the candidate pool at every iteration, execute its candidates from the same restored simulator state, and then score that pool with every surrogate. This separates two roles: the *generator* produced the pool through its earlier updates; the *scorer* ranks the saved pool now. We call this a *crossed design* because each of the four generators is paired with each of the four scorers, giving all 16 combinations. A column follows one scorer across pool sources. A row holds the pool fixed and compares scorers. The matrix describes proposal disagreement associated with generators, scorers, and their pairings. It does not assign causal effects to those components.

The intervention tests a consequence that rescoring alone cannot establish. At the first update, we fit one proposal from model-ranked elites and another from environment-ranked elites in the same pool. Both branches then continue with the original scorer and matched random samples. This *one-update splice* changes how later candidates are generated through one earlier decision. Its endpoint is the realised cost of the sequence selected after the remaining CEM iterations. Figure 1 shows the audit and intervention together.

2 Related work

Model evaluation has long depended on the decisions that use the model. Value-aware model learning weights errors through downstream value functions [Farahmand et al., 2017]. DeepMDP connects reward and latent transition prediction to the properties of the learned representation [Gelada et al., 2019]. Value equivalence compares models through their effects on specified value functions and policies [Grimm et al., 2020]. Lambert et al. [2020] examine the mismatch between one-step prediction likelihood and downstream control performance. These works give distinct reasons to

evaluate a model through its intended use. We hold the predictors fixed and examine the finite candidate decisions made by CEM.

Yu et al. [2026] make a broader case for evaluating world models by interventional action fidelity, policy ranking, and optimisation utility. Their proposed reset-matched action branches are directly relevant to our environment reference: every candidate starts from the same physical state. Bhamidipaty et al. [2026] define model exploitation through a reversal of policy preference between the model and the environment, and analyse when such reversals can occur. Our elite and ranking measurements examine finite candidate disagreements; they do not estimate the worst-case exploitability of a model over a policy class.

The closest CEM diagnostic is the CEM-stage Spearman analysis of Wang et al. [2026]. It measures agreement between model and realised cost rankings as search moves from initial candidates towards concentrated and elite populations. This already evaluates the model on queries made during search. Our crossed design compares a scorer’s own pools with pools generated by other surrogates. Holding the scorer fixed tests how its disagreement varies with pool source. Holding the pool fixed compares scorers without changing candidates. We also compare the means and widths fitted from the two elite sets, because these parameters determine the next queries. A rank correlation alone does not specify that refit.

Li et al. [2026] execute all candidates in saved proposal pools to distinguish failed selection from the absence of a feasible candidate. Their adjacent-set method then reconstructs an action sequence from a neighbourhood of low-cost candidates. We use the same basic control of executing candidates from a common initial state. Our intervention changes an earlier elite refit and follows its consequence through later pools. This tests how the proposal history changes the candidates available at final selection.

Several recent methods change components that our audit keeps fixed. Chahe et al. [2026] train a cost-ranking objective that penalises increasing action perturbations. Huang et al. [2026] add a trajectory-level cost to endpoint scoring, and Singh [2026] change the planning objective without retraining the predictor. Sommer and Schilling [2026] learn a plan-improvement operator through imagined rollouts. These methods change candidate scores or proposal updates. Applying the crossed comparison before and after such a change would test which measured disagreements it reduces.

3 Auditing a CEM update

The audit begins with a fixed planning problem: one saved simulator state and its matching Dreamer context. We run CEM from that context, save each sampled pool, and compare the decisions obtained from model predictions with those obtained from executing the same candidates. Restoring the simulator state before every execution prevents one candidate from changing the starting point of another. Realised cost supplies the reference ordering within that pool.

Candidates and elites. The proposal is a diagonal Gaussian: each action coordinate has its own mean and standard deviation, with no fitted correlations between coordinates. Here a candidate contains 15 actions, each with six coordinates. We flatten this 15×6 array by listing its entries in one vector of dimension $d = 90$. The index r identifies an action coordinate at one planning step; flattening does not discard or average any actions.

Write the current proposal mean and standard deviation as μ and σ . A candidate j is sampled as

$$A_j = \text{clip}(\mu + \sigma \odot \epsilon_j), \quad \epsilon_j \sim \mathcal{N}(0, I_d), \quad j = 1, \dots, 256. \quad (1)$$

The product $\odot$ scales each noise coordinate by its proposal standard deviation. Clipping enforces the action bounds after sampling. All later comparisons use these bounded candidates, including the environment refit.

The scorer f assigns a predicted cost to each sequence. Its 32 lowest-cost sequences form E^f . The 32 sequences with lowest realised cost form the environment elite set E^E . Both sets contain candidates from the same pool; the environment reference supplies their realised ordering.

Elite precision records the fraction of model elites also present in the environment elite set:

$$p = \frac{|E^f \cap E^E|}{32}, \quad p_{\text{adj}} = \frac{p - 32/256}{1 - 32/256}. \quad (2)$$

Two independently selected random sets of size 32 have expected overlap four, giving expected precision $32/256 = 0.125$. Subtracting that baseline and rescaling gives zero for chance overlap and one for identical sets. Negative values indicate less overlap than the chance baseline. This measurement focuses on which candidates CEM retains.

Pairwise order agreement covers the rest of the pool. It counts the fraction of candidate pairs for which the scorer preserves the environment ordering, excluding pairs tied in realised cost. Selected-candidate regret instead subtracts the best realised cost in the pool from the realised cost of the scorer’s top candidate. It measures the cost of selection within the available pool. Neither measurement says what candidates the next proposal will produce. We retain Spearman agreement, Jaccard elite overlap, and alternative elite cuts as supporting records.

3.1 Comparing the two proposal refits

The elite indices alone do not determine the size of an update error. Two disjoint elite sets could contain very similar actions and fit similar proposals. Conversely, a small change in elite membership could move the proposal if the changed sequences lie far apart. We therefore apply CEM’s actual refit to both elite sets.

For each coordinate, CEM fits the elite mean and standard deviation, flooring the standard deviation at 0.05. The model elites produce (μ^f, σ^f) ; the environment elites produce (μ^E, σ^E) . These are *next-proposal* parameters. The symbol σ without a superscript belongs to the *current* proposal that generated the pool.

The mean distance D_μ compares the centres of the two refits. We first divide each difference by the corresponding current proposal width, then take a root mean square over the 90 coordinates:

$$D_\mu = \sqrt{\frac{1}{d} \sum_{r=1}^d \left(\frac{\mu_r^f - \mu_r^E}{\sigma_r} \right)^2}. \quad (3)$$

$D_\mu = 1$ means a root-mean-square separation of one current proposal standard deviation. The denominator makes the scale explicit: the same raw mean difference occupies more of the remaining search region when CEM has already narrowed its proposal. A rise in D_μ need not mean that the raw action difference rises.

The width comparison uses a different convention. For each coordinate, we compare the two fitted standard deviations by their ratio:

$$D_{\log \sigma} = \sqrt{\frac{1}{d} \sum_{r=1}^d \log^2 \left(\frac{\sigma_r^f}{\sigma_r^E} \right)}. \quad (4)$$

A doubling and a halving have equal log magnitude. Equal fitted widths give zero, regardless of how narrow they are. Our pre-specified distance combines the squared mean and log-width terms with equal weight:

$$D = \sqrt{\frac{D_\mu^2 + D_{\log \sigma}^2}{2}}. \quad (5)$$

$D = 0$ exactly when the two elite refits produce the same diagonal Gaussian. Its two terms measure different kinds of disagreement, so their trajectories must be read separately. A fall in D can come from widths becoming more similar in ratio even as the means separate in current-proposal units.

We also compute a width difference in those same current-proposal units:

$$D_\sigma = \sqrt{\frac{1}{d} \sum_{r=1}^d \left(\frac{\sigma_r^f - \sigma_r^E}{\sigma_r} \right)^2}, \quad D_W = \sqrt{D_\mu^2 + D_\sigma^2}. \quad (6)$$

D_W is the 2-Wasserstein distance between the two diagonal Gaussian refits after dividing coordinates by the current proposal widths, normalised by $\sqrt{d}$. For diagonal Gaussians, the squared distance is the sum of squared mean and standard-deviation differences. Equation 6 applies that expression coordinatewise. Comparing D with D_W checks whether the interpretation depends on using width ratios or width differences. Neither distance alone measures candidate ordering or realised cost.

The first action. MPC executes only the first of CEM’s 15 planned actions before replanning. A distance averaged over the full plan can conceal disagreement in that action. We therefore also compute D using only its six coordinates.

We also compare the first actions of the scorer’s lowest-cost sequence and the environment’s lowest-cost sequence. Their root-mean-square gap is reported in raw action units and after division by the current first-action proposal widths. The raw gap records separation on the bounded action scale. The normalised gap records separation relative to the region CEM is still searching. A shrinking proposal can make these quantities move in opposite directions.

Clipping and the standard-deviation floor provide two further checks on that interpretation. We record proposal width, floor occupancy, the clipping rate, and how far clipping displaces the sampled pool mean. These records distinguish contraction of the proposal from changes introduced by its bounds.

4 Crossed audit and one-update splice

Crossed audit. Every surrogate receives the same actor-centred proposal and random samples at iteration zero. Later pools inherit the elite decisions made along each trace. The generator g produced a pool through those preceding decisions; the scorer f ranks that fixed, realised pool.

All generator traces use the same standard-normal samples at every iteration. Their bounded candidates match at iteration zero because their proposal parameters also match. Later candidates differ when generators fit different means or standard deviations. Common samples remove independent Monte Carlo variation from the row comparison.

Scoring every generator pool with every surrogate gives 16 distances at iteration i , written D_{gfi} . Each row reuses one candidate pool across four scorers. Each column follows one scorer across four generated pools. The usual model-driven trace occupies the diagonal, where $g = f$.

Each scorer has one baseline distance D_{f0} on the common iteration-zero pool. Alongside absolute distances, we show the change $D_{gfi} - D_{f0}$. Subtracting this baseline separates a scorer’s initial level of disagreement from the change observed after generator-specific updates. It does not remove the dependence on the distance definition.

The matrix supports comparisons that a diagonal trace cannot provide. Suppose one scorer produces a larger distance on pools from all four generators. That appears as a high-distance column and associates the disagreement with current scoring. Suppose instead that all four scorers produce larger distances on one generator’s pool. That appears as a high-distance row and associates the disagreement with the queried actions. The physical state, task cost, and candidate count remain fixed in both comparisons.

A particular cell may also depart from these row and column patterns. For cell (g, f) , we subtract its row and column means and restore the matrix mean:

$$\text{interaction} = \text{cell distance} - \text{row mean} - \text{column mean} + \text{matrix mean}. \quad (7)$$

We write this two-way residual as I_{gfi} . The matrix mean corrects for subtracting the overall level twice. On the diagonal, $g = f$: the same surrogate generated and scored the pool. A positive I_{gfi} means that its own-pool distance exceeds the value predicted by adding its row and column differences to the matrix mean. It describes a pairing discrepancy beyond those two averages.

We summarise row, column, and residual variation by their shares of the matrix’s sum of squares. Each share is calculated within one task-seed matrix, then averaged over the six seeds for that task. Decomposing an already averaged matrix would discard between-seed differences in its structure. These percentages describe the four fixed families; they are not population variance estimates. Appendix B gives the exact calculation.

Row differences alone do not show that an earlier elite decision caused harm. Rescoring measures disagreement on the inherited pool; an intervention is needed to test an update’s consequence. The distance matrix also does not measure the lowest cost available in a pool. Even perfect agreement with the environment elites can occur when every candidate has high realised cost. We retain realised candidate costs to distinguish that coverage problem from disagreement within the pool.

At iteration zero, the four generator labels must point to identical candidate IDs and environment costs. We verify this before analysing later pools. Row differences can appear only after the first model-ranked update.

One-update splice. To test an elite decision’s consequence, we fork the search at iteration zero. Every generator has the same proposal and candidates at this point. Using this common starting pool allows the two branches to differ in one elite set without inheriting an earlier difference in search history. Branch M fits its next proposal from model elites; branch E fits from environment elites. Both use the same mean and standard-deviation refit, including the width floor.

From iteration one, the original surrogate scores both branches. Matched standard-normal samples are scaled and shifted by the two inherited proposal states. All subsequent elite sets are selected using the original surrogate. We reuse the model branch from the audit and execute the new candidates in branch E .

After each iteration, the controller retains the lowest predicted-cost sequence seen so far, including candidates from earlier pools. This *incumbent* is the sequence that supplies its first action if planning stops at that point. We evaluate each branch’s incumbent after iterations one, two, and three. The primary endpoint is the branch- E realised cost minus the branch- M realised cost after iteration three, the final CEM iteration. A negative contrast favours the environment-elite branch. The intermediate endpoints show whether the contrast changes direction during the remaining search; they cannot replace the fixed final endpoint.

We also record the lowest realised candidate cost reached by each branch. A branch can contain a lower-cost candidate without selecting it, so we compare the best available cost with the incumbent’s realised cost.

Only the first update is replaced; later-stage interventions are not tested. Environment elites require realised costs for every candidate, so the splice is an oracle diagnostic rather than a deployable planner. Its endpoint measures a selected sequence’s finite-horizon cost, not closed-loop episode return.

5 Experimental setting

The crossed audit contains 12 *task-seed units*: six Walker seeds and six Cheetah seeds. Each unit has an independently trained Dreamer backbone and actor [Hafner et al., 2020]. We freeze both. The backbone maps observation history to a 1280-dimensional context c_t ; the actor supplies the initial CEM mean. Four separately fitted surrogates propagate c_t under proposed actions.

Each surrogate maps the current context into a feature space, propagates those features for 15 action-conditioned linear transitions, and decodes every predicted state back to Dreamer’s context space. The rollout receives no intermediate Dreamer states. Training minimises free-running decoded-context error at horizons 1, 3, 5, 8, and 15, plus reward and continuation losses. A separate validation split selects the checkpoint.

The four families change the feature map φ and its decoder. *Identity* uses $\varphi(c) = c$. *Learned nonlinear* uses a trainable multilayer perceptron. *Anchored nonlinear* concatenates trainable affine and nonlinear features. *Random nonlinear* uses fixed random nonlinear features. The three compressed families have width 256 and share the decoder architecture, auxiliary heads, optimiser, data, and training budget. Identity retains the 1280-dimensional context and is not width matched. All four use the same action-conditioned transition form.

We use `walker_walk` and `cheetah_run` from the DeepMind Control Suite [Tassa et al., 2018, Tunyasuvunakool et al., 2020, Todorov et al., 2012]. Seeds 0–2 form the original six-unit cohort used to decide which splice to run. The follow-up rule required a pattern in at least two of three seeds on each task, retained after omitting any one saved state. This limited dependence on a single seed or decision point. The original questions and rule were fixed before inspecting their outcomes. Later analyses of distance components, raw action gaps and clipping explain the results without changing the selected pairing. The rule governed follow-up computation; it was not a significance test. Appendix A gives its exact criteria and chronology.

Seeds 3–5 were fixed before inspecting their audit outcomes to increase the precision of the same comparisons. They use the same state-selection, replay, model and CEM settings and cannot change

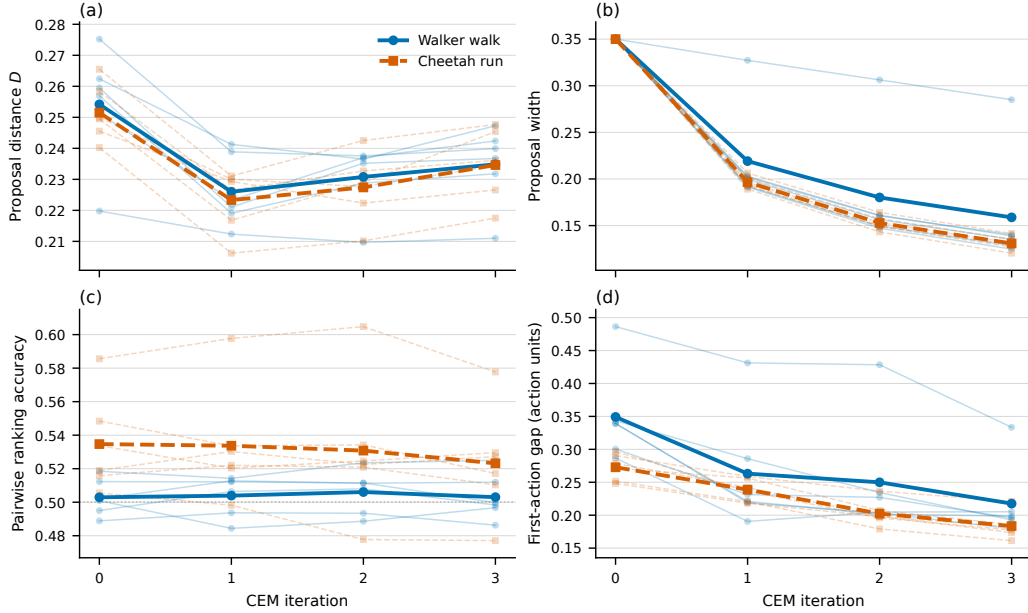


Figure 2: Diagonal audit measurements over four CEM iterations. Thin lines show six task-seed units per task after averaging the four surrogates; heavy lines show task means. The model bank, candidate count, elite cut, and horizon are fixed. From iteration zero to three, D and the raw first-action gap decrease with proposal width. Pairwise order agreement stays near chance on Walker and declines on Cheetah. The dotted line in (c) marks chance agreement.

the original selection. We also fixed the selected splice and its decision rule before inspecting these six units’ splice costs. This held-out test checks whether the same intervention lowers final selected-sequence cost on new units. It does not compare other families as intervention targets. We report the two cohorts separately as well as the combined task means.

Candidate scoring begins at a saved context and predicts 15 steps. A fixed ridge readout maps decoded contexts to the task variables in the cost. Walker tracks a fixed goal; Cheetah tracks the frozen actor’s reference trajectory. Predicted and realised costs use the same task variables, reference targets and cost weights. Each CEM run has four iterations, 256 candidates, and 32 elites. The proposal width starts at 0.35 and has a floor of 0.05. Surrogate parameters stay frozen and training-time stochasticity is disabled. Appendix Table 9 lists the remaining model, training, readout, and task-cost settings.

For each unit, we export 64 simulator-state/context pairs from the frozen actor. The audit uses one state from each of four pre-selected episode bins after a replay check. The bins spread decision points across an episode without selecting states for their audit outcome. We restore the exact state before every candidate execution and verify repeated costs. Each generator trace uses the same standard-normal samples at every iteration; differences in its proposal parameters produce differences in its later bounded actions.

The complete crossed audit executes 196,608 candidates; the original six-unit cohort accounts for 98,304. Each pool is executed once and then scored by all four surrogates, so the off-diagonal cells add model evaluations without adding simulator rollouts. We average states within each task-seed unit before comparing seeds. The diagonal trajectories also average the four model families; crossed matrices retain every generator-scorer cell. Candidate sequences, matrix cells, and saved states are repeated measurements within a unit, not independent replicates.

Measurement	Walker		Cheetah	
	Iter. 0	Iter. 3	Iter. 0	Iter. 3
Mean distance D_μ	0.200	0.288	0.184	0.292
Log-width distance $D_{\log \sigma}$	0.293	0.165	0.303	0.156
Linear-width distance D_σ	0.165	0.147	0.160	0.138
Whitened distance D_W	0.260	0.324	0.244	0.324
First-action proposal distance D	0.349	0.326	0.261	0.252
First-action gap (action units)	0.349	0.218	0.273	0.184
First-action gap (proposal SD)	0.998	1.321	0.779	1.258

Table 1: Components of the proposal refit and first-action disagreement. Values average diagonal model traces and states within task-seed units, then the six seeds per task. The candidate count, elite cut, and horizon remain fixed. Fitted means separate in current-proposal units as width disagreement falls. Raw first-action gaps shrink, but width-normalised gaps grow. These quantities cannot be read as interchangeable measures of elite agreement.

6 Results

6.1 The proposal distance falls without better candidate ranking

The pre-specified proposal distance decreases from iteration zero to three in all original and extension units. In the combined 12-unit result, Walker’s task mean falls from $D = 0.254$ at iteration zero to 0.235 at iteration three; Cheetah falls from 0.252 to 0.235 (Figure 2). Every task-seed contrast is negative. They range from -0.035 to -0.009 on Walker and from -0.023 to -0.004 on Cheetah.

The two components of D move in opposite directions (Table 1). The mean term D_μ rises on both tasks, so the two elite refits separate relative to the current proposal width. The log-width term falls. Its reduction accounts for the lower D despite the rise in mean disagreement. The alternative distance D_W rises by 25% on Walker and 33% on Cheetah. A falling D records a reduction in its ratio-based width term. It does not establish that CEM retains more of the environment elites or selects a lower-cost sequence.

The first-action gap gives the same warning about scale. The scorer’s selected action and the environment’s selected action become closer in raw action units, yet farther apart relative to the current proposal width. CEM’s search region contracts faster than their separation. This scale dependence is already present at the first planned step.

At iteration three, only 1.2% of Walker coordinates and 1.4% of Cheetah coordinates are at the width floor. Most widths therefore contract without reaching the minimum. The mean displacement from clipping also decreases. Table 8 gives its endpoints, and Figure 8 records clipping rates along the traces.

Pairwise order agreement changes from 0.503 to 0.503 on Walker and from 0.535 to 0.523 on Cheetah. Chance-adjusted elite precision changes from 0.000 to -0.001 on Walker and from 0.071 to 0.040 on Cheetah. On the original seeds 0–2, the neighbouring elite cuts keep task-mean adjusted precision below 0.10. Figure 7 reports these cuts.

6.2 Scorer variation dominates on Cheetah

At iteration three, scorer columns account for 66.2% of the descriptive sum of squares on Cheetah; pool-source rows account for 10.0% and pairings for 23.8%. Walker is mixed: the corresponding fractions are 28.7%, 31.6%, and 39.7% (Figure 3). Scorer variation exceeds pool-source variation in 6 of 6 Cheetah seeds, compared with 2 of 6 Walker seeds. Thus the Cheetah column structure is present within individual seeds, rather than appearing only after their matrices are averaged.

Random nonlinear has a positive own-pool residual in all six original units. Its distance on its own generated pool exceeds the additive value from its row and column averages. Every one-state omission preserves the sign. Appendix A gives the seed-level residuals, and Table 4 gives the omitted-state results. In the combined audit, the residual is positive in 5 of 6 Walker seeds and 5 of 6 Cheetah seeds.

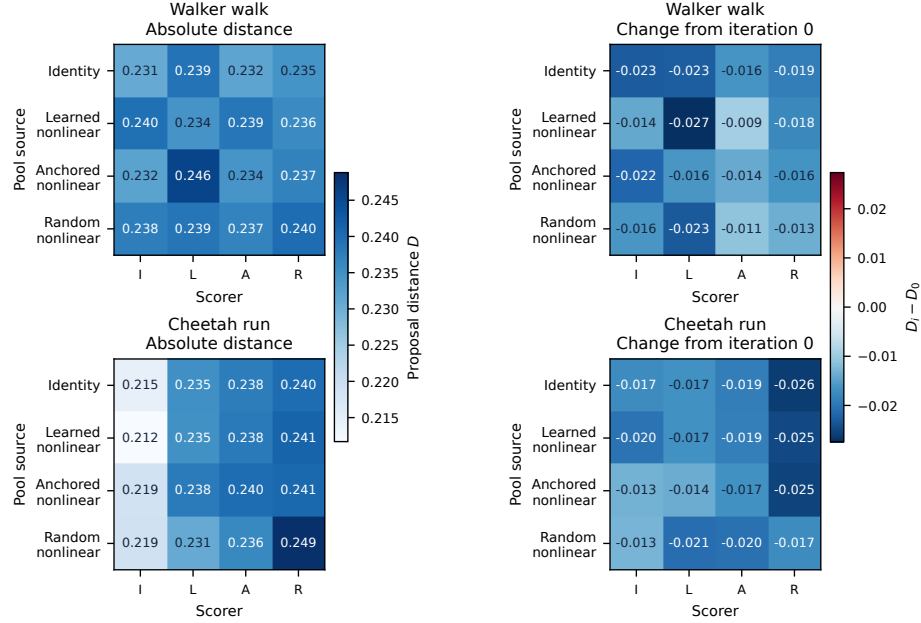


Figure 3: Final-iteration crossed proposal distance, averaged over six seeds per task. The attribution percentages instead average the six within-seed decompositions described in Section 4. A row fixes the surrogate that generated a pool; each column rescores that executed pool with Identity, Learned nonlinear, Anchored nonlinear, or Random nonlinear. The right panels subtract each scorer’s value on the common initial pool. Cheetah has greater variation between scorers than between pool sources in all six seeds. Walker divides variation across pool source, scorer, and pairing.

The original decision rule therefore selected Random nonlinear for the splice. No pool-source row met the cross-task criterion.

6.3 Replacing the first elite set lowers final selected-sequence cost

The pre-specified Random nonlinear splice replaces the first model-elite update with the environment-elite update and then returns both branches to the same surrogate. Before averaging within a task-seed unit, we divide each state-level cost contrast by the difference between the 90th and 10th percentiles of realised cost in its shared initial pool. This is the pool’s *interdecile range*. A value of -0.5 , for example, means that the environment-elite branch selects a sequence whose realised cost is lower by half of that initial-pool range.

The final contrast is negative in all 6 selection units. The task means are -0.405 on Walker and -0.912 on Cheetah.

The direction repeats in all 6 held-out units. Their task means are -0.253 on Walker and -0.620 on Cheetah. The fixed cross-task rule survives all 4 one-state omissions. At least two of three seeds per task retain a negative contrast after each omission. Omitting one state makes Walker seed 5 slightly positive. Table 6 records the exception. Across both cohorts, all 12 task-seed contrasts are negative; the combined descriptive means are -0.329 and -0.766 (Figure 4).

The best realised candidate also has lower cost in every task-seed unit. Its combined contrast is -0.053 initial-pool ranges on Walker and -0.442 on Cheetah. The Cheetah splice therefore reaches pools containing lower-cost sequences. Walker’s best-candidate change is smaller than its selected-cost change, indicating that coverage alone does not describe the selected-cost contrast. Later model selection also differs between the branches. Several held-out units have a positive contrast after iteration one. Replacing the first elite set does not improve every intermediate endpoint; its benefit appears at the fixed final endpoint.

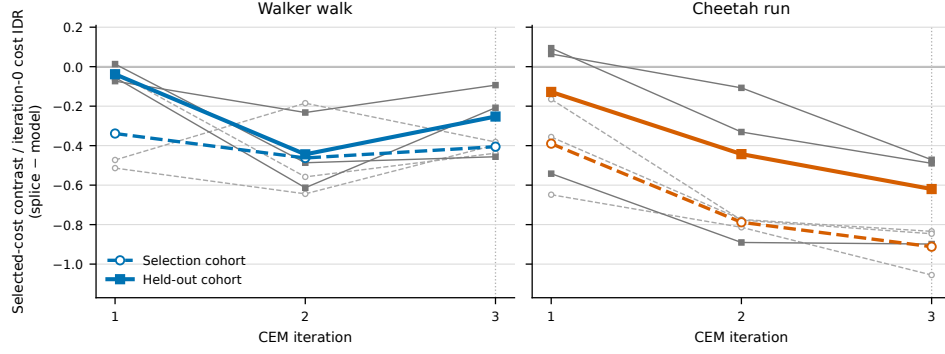


Figure 4: Effect of replacing the first Random nonlinear elite set on selected cost over iterations one to three. Both branches share the initial pool, later scorer, and base random samples; only the first elite set differs. Environment-elite minus model contrasts are scaled by the initial pool’s realised-cost interdecile range. Dashed circles show selection units, solid squares show held-out units, and lines show cohort means. Every final contrast is negative, including all six held-out units; intermediate contrasts can be positive. Table 5 reports all 12 task-seed trajectories.

7 Discussion and limitations

A proposal audit needs an explicit reference scale. Mean separation in raw action units answers a different question from separation relative to the remaining search width. Width ratios answer another. Reporting those components alongside elite recovery prevents contraction alone from being read as improved decision agreement. Realised selected-sequence cost is still needed to test whether an update changes the outcome of search.

The crossed comparison guides the next controlled test. On Cheetah, a scoring change can be tested across several fixed pool sources, because the observed differences follow scorer columns in every seed. On Walker, a score-only account omits row and pairing variation. A proposal constraint could be tested with the scorer held fixed, or a scoring change could be tested on the same saved pools. The matrix makes these comparisons available without additional off-diagonal simulator executions. It does not choose a remedy or guarantee that one will work.

The Random nonlinear splice shows that one elite decision can affect the selected cost after three further CEM iterations. Its direction repeats on units held out from pairing selection. Other surrogates might also benefit from an environment-elite update: we have not compared splices across all four families. The result therefore establishes the consequence of the selected update, without establishing that the residual identified the best intervention target.

The experiment covers two tasks, six independently trained Dreamer backbones per task, four states per backbone, one horizon, and one CEM configuration. The shared transition form controls part of the model comparison; other dynamics classes may produce different matrices. Identity also retains a wider context than the three compressed families, so family differences cannot be assigned to feature type alone. The four states sample a restricted part of each actor’s experience, and state omissions check dependence on those points rather than coverage of all reachable states.

The audit requires a resettable environment and execution of every candidate. It is suited to diagnosing a model-planner loop in simulation. The splice uses information unavailable to an ordinary deployed planner, and lower finite-horizon selected-sequence cost does not establish higher closed-loop return. A method motivated by the audit must be evaluated without that oracle and under receding-horizon control.

Conclusion. A world model used by CEM determines both the ordering of current candidates and the distribution of later queries. Crossed rescoring describes these roles separately; a matched one-update splice tests whether an elite decision changes the final selected-sequence cost.

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A Follow-up rule and experiment chronology

The original audit used Walker and Cheetah seeds 0–2, with four saved states, four surrogate families, and four CEM iterations per task-seed unit. Its gate was a rule fixed before outcome inspection for deciding whether to extend the audit and run an intervention. It was not a significance test.

Original questions. The first question tested whether the family-mean diagonal distance D increased from iteration zero to three. The second tested whether pool source or an own-pool pairing produced a replicated pattern at iteration three. A pool-source pattern required at least three of four scorer columns to name the same highest-distance generator within a task; that generator could differ between tasks. A pairing pattern required the same surrogate to have a positive diagonal residual I_{ff3} on both tasks.

A qualifying pattern needed the stated direction on Walker and Cheetah, at least two of three seeds per task, and the same outcome after each of the four one-state omissions. The omissions recomputed task-seed summaries from the remaining three states. This rule limited reliance on one seed or one saved state. It did not estimate the probability of replication on a new task or provide a multiple-testing correction. Only the second question could trigger the one-update splice.

The original measurements included D , pairwise order agreement, and elite recovery. Selected-candidate regret, alternative elite cuts, and downstream family ordering did not enter the gate. We specified the change-from-baseline matrices, descriptive sums of squares, random-elite reference and D_W after inspecting the Walker results, before inspecting the completed Cheetah summaries. The mean/width decomposition, raw-action gap, floor occupancy and clipping displacement followed inspection of the six-unit pilot and preceded completion of the extension cohort. These analyses could not change the original rule or the selected family.

Gate outcome. D decreased, contrary to the original increasing-distance hypothesis. No pool-source pattern qualified under the seed and state-omission checks. Random nonlinear’s positive own-pool residual qualified and selected its splice. At iteration three, I_{ff3} is 0.0023, 0.0021, and 0.0035 in Walker seeds 0, 1, and 2; the Cheetah values are 0.0090, 0.0018, and 0.0050. Every one-state omission preserves their positive sign (Table 4). No splice cost was used for this selection.

Precision extension and held-out splice. Seeds 3–5 were fixed as an audit precision extension before their audit outcomes were inspected. They use the same model, state-selection, replay, and CEM settings. They do not reopen the original gate. After Random nonlinear was selected on seeds 0–2, the same first-update splice and its held-out decision rule were fixed before inspecting seeds 3–5 splice costs. The rule requires a negative final selected-cost contrast in at least two of three seeds on each task, including after every one-state omission. All six full-state contrasts are negative. The rule also passes under every omission, although omitting bin 3 makes Walker seed 5 slightly positive (Table 6).

The first update is the intervention point because all traces then share the same pool. Iteration three is the primary endpoint because it ends the fixed four-iteration search. Intermediate contrasts are reported at iterations one and two, including positive contrasts. We did not select an intervention stage or endpoint by its realised gain. This experiment tests the selected update’s consequence and its repetition on held-out units; it does not compare the four families as intervention targets.

B Descriptive attribution calculation

At iteration i , the state-averaged task-seed matrix contains $K = 4$ generator rows and $K = 4$ scorer columns. Write its row, column, and grand means as $\bar{D}_{g \cdot i}$, $\bar{D}_{\cdot fi}$, and $\bar{D}_{\cdot \cdot i}$. Define the row and column deviations from the grand mean by

$$R_{gi} = \bar{D}_{g \cdot i} - \bar{D}_{\cdot \cdot i}, \quad C_{fi} = \bar{D}_{\cdot fi} - \bar{D}_{\cdot \cdot i}. \quad (8)$$

The remaining cell residual is $I_{gfi} = D_{gfi} - \bar{D}_{\cdot i} - R_{gi} - C_{fi}$, equivalent to Equation 7. The sums of squares are

$$SS_{\text{row}} = K \sum_g R_{gi}^2, \quad SS_{\text{col}} = K \sum_f C_{fi}^2, \quad (9)$$

$$SS_{\text{res}} = \sum_{g,f} I_{gfi}^2, \quad SS_{\text{tot}} = \sum_{g,f} (D_{gfi} - \bar{D}_{\cdot i})^2. \quad (10)$$

They satisfy $SS_{\text{tot}} = SS_{\text{row}} + SS_{\text{col}} + SS_{\text{res}}$. For each component x , we compute $100 SS_x / SS_{\text{tot}}$ within each task-seed matrix, then average the six shares for a task. We do not decompose the six-seed mean matrix to obtain the reported percentages. The families are fixed test models, so the shares describe this matrix rather than population variance components or causal effects.

C Seed-level crossed matrices

Metric	Walker		Cheetah	
	Iter. 0	Iter. 3	Iter. 0	Iter. 3
Proposal distance D	0.254	0.235	0.252	0.235
Proposal width	0.350	0.159	0.350	0.131
Whitened distance D_W	0.260	0.324	0.244	0.324
First-action gap (action units)	0.349	0.218	0.273	0.184
First-action gap (proposal SD)	0.998	1.321	0.779	1.258
Pairwise order agreement	0.503	0.503	0.535	0.523
Adjusted elite precision	0.000	-0.001	0.071	0.040

Table 2: Task means at the common initial pool and final CEM iteration. The horizon, candidate count, elite cut, and four-surrogate average are fixed. D and the raw first-action gap decrease; D_W and the width-normalised action gap increase. Pairwise order agreement stays near chance on Walker and declines on Cheetah. Chance-adjusted elite precision does not improve.

Figures 5 and 6 retain each of the six original gate units before task averaging. The main figures add the six extension units. Rows and columns follow Identity, Learned nonlinear, Anchored nonlinear, and Random nonlinear; the axes use the first word of each name.

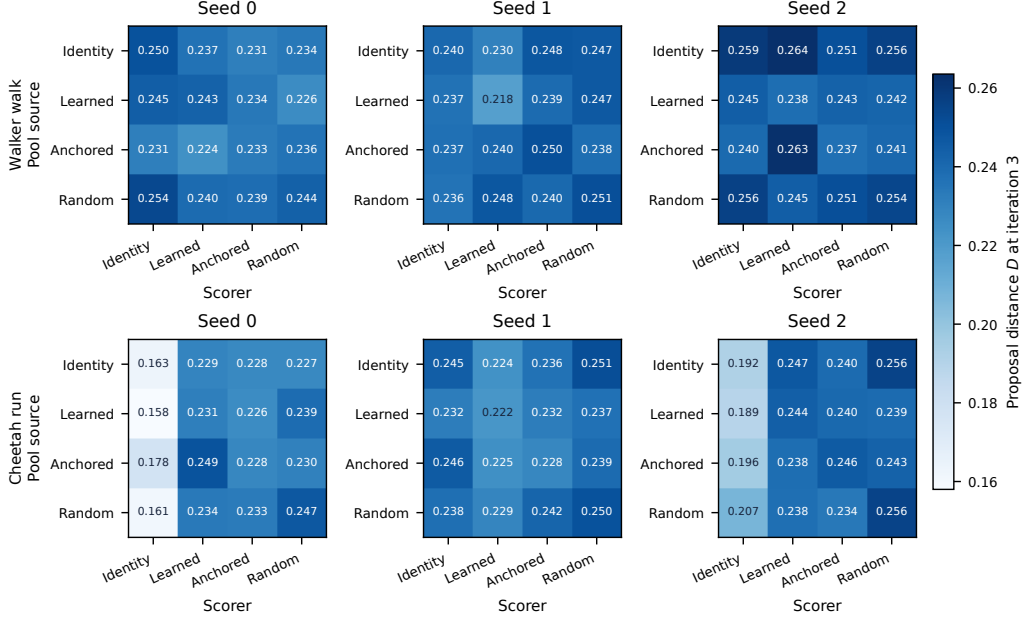


Figure 5: Final-iteration crossed proposal distance for the six seed 0–2 gate units. A row fixes the generator pool and columns rescore its executed candidates; candidate sets are never pooled across seeds. Cheetah is scorer-dominated in all three gate units. Identity has the lowest column mean in seeds 0 and 2; Learned nonlinear has the lowest in seed 1.

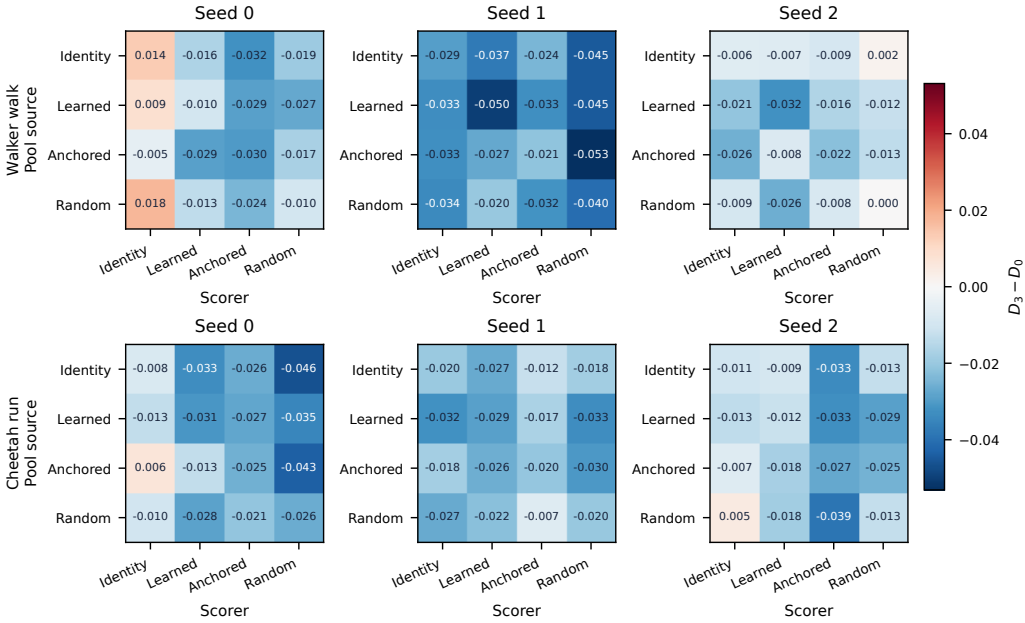


Figure 6: Change in crossed proposal distance for the six gate units. Each cell holds its scorer fixed and subtracts that scorer's value on the common iteration-zero pool. Most cells are negative at iteration three. This comparison records the ratio-based distance D and does not determine candidate order agreement.

D Leave-one-state-out checks

Each task–seed unit contains four saved states drawn from fixed replay time bins. We repeat the gate and crossed attribution after omitting one bin at a time. We average the remaining three states within each task–seed unit before comparing seeds.

Omitted bin	Task	Seed 0	Seed 1	Seed 2	n_+	Pass
All states	Walker walk	-0.009	-0.035	-0.015	0	no
All states	Cheetah run	-0.023	-0.022	-0.016	0	no
−1	Walker walk	-0.011	-0.027	-0.011	0	no
−1	Cheetah run	-0.030	-0.025	-0.019	0	no
−3	Walker walk	-0.000	-0.040	-0.020	0	no
−3	Cheetah run	-0.020	-0.017	-0.025	0	no
−5	Walker walk	-0.018	-0.035	-0.021	0	no
−5	Cheetah run	-0.018	-0.028	-0.011	0	no
−7	Walker walk	-0.007	-0.039	-0.008	0	no
−7	Cheetah run	-0.023	-0.019	-0.007	0	no

Table 3: Leave-one-state-out check for the proposal-distance gate. Each column is a task–seed contrast after averaging model families; n_+ counts contrasts above zero. The task, seeds, CEM settings, and remaining three states are fixed within an omission. Every omission preserves a negative iteration-three minus iteration-zero contrast on both tasks.

Omitted bin	Walker support	Cheetah support	Cross-task self interaction	Support	Interaction
All states	none	none	Identity; Random nonlinear	no	yes
−1	none	none	Random nonlinear	no	yes
−3	none	none	Identity; Random nonlinear	no	yes
−5	none	none	Identity; Random nonlinear	no	yes
−7	Random nonlinear	none	Random nonlinear	no	yes

Table 4: Leave-one-state-out check for the two attribution triggers. A support trigger requires at least three scorer columns to select the same highest-distance generator within each task. An interaction trigger requires the same surrogate’s own-pool residual to be positive on both tasks. Random nonlinear passes under all four omissions; no generator passes the support trigger.

E Seed-level elite-splice results

The main paper plots the paired normalised contrasts. Table 5 retains the fixed raw endpoint and every task–seed trajectory for the selection and held-out cohorts. The environment-elite branch differs only in the first update; iterations one to three again use Random nonlinear scores in both branches.

Cohort	Task	Seed	Δ_1/s_0	Δ_2/s_0	Δ_3	Δ_3/s_0	$\Delta_3^{\text{best}}/s_0$
Selection	Walker walk	0	-0.029	-0.558	-1409.812	-0.438	-0.018
		1	-0.514	-0.644	-1304.868	-0.396	-0.052
		2	-0.473	-0.185	-801.631	-0.381	-0.049
	Cheetah run	0	-0.356	-0.774	-3124.093	-0.834	-0.379
		1	-0.649	-0.814	-2880.493	-1.055	-0.431
		2	-0.165	-0.777	-1852.621	-0.846	-0.494
Held-out	Walker walk	3	-0.055	-0.613	-710.564	-0.208	-0.139
		4	+0.013	-0.487	-1776.227	-0.456	-0.051
		5	-0.073	-0.232	-760.797	-0.093	-0.008
	Cheetah run	3	+0.094	-0.331	-1345.455	-0.489	-0.528
		4	-0.542	-0.890	-3394.181	-0.898	-0.500
		5	+0.064	-0.107	-1125.465	-0.471	-0.318

Table 5: Paired realised-cost contrasts for the 12 Random nonlinear splice units. Seeds 0–2 selected the pairing without using splice costs; seeds 3–5 test the fixed intervention without using their splice outcomes for selection. Both branches hold the initial pool, later scorer, and base samples fixed. Δ_i is environment-elite branch minus model branch after iteration i , Δ_3^{best} compares the best realised candidates reached, and s_0 is the realised-cost interdecile range in the shared initial pool. Every iteration-three selected-cost contrast is negative.

Omitted bin	W3	W4	W5	C3	C4	C5	Rule
None	-0.208	-0.456	-0.093	-0.489	-0.898	-0.471	Pass
1	-0.228	-0.523	-0.250	-0.548	-1.003	-0.302	Pass
3	-0.170	-0.381	+0.001	-0.478	-0.651	-0.484	Pass
5	-0.164	-0.315	-0.006	-0.542	-1.025	-0.561	Pass
7	-0.269	-0.606	-0.119	-0.391	-0.915	-0.539	Pass

Table 6: Leave-one-state-out check for the held-out splice. Entries are final normalised selected-cost contrasts for Walker (W) and Cheetah (C). The family, CEM settings, and remaining three states are fixed. Omitting bin 3 makes Walker seed 5 slightly positive, but the pre-specified rule still passes with two negative Walker seeds and three negative Cheetah seeds. All other omissions retain six negative contrasts.

Audit pattern	Reading and next check
Scorer columns dominate	Test a score change or ranking loss against realised pairwise order and elite recovery.
Pool-source rows dominate	Test a proposal constraint or actor-anchored sampling.
Positive diagonal residual	Run a targeted model-planner intervention.
Metrics split as width contracts	Report width, ranking, and action-level fidelity.

Table 7: Matrix patterns and possible controlled follow-ups. The candidate pool is fixed across scorer columns, and the scorer is fixed down a column. Row, column, and diagonal patterns therefore direct the next test to a different component; none establishes that a proposed response will work.

F Elite-set size

The primary CEM update uses 32 elites. We recompute the final diagonal measurements with 8, 16, and 64 elites from the saved candidate pools for the original seeds 0–2. These are alternative refits of existing pools; they do not generate new CEM traces.

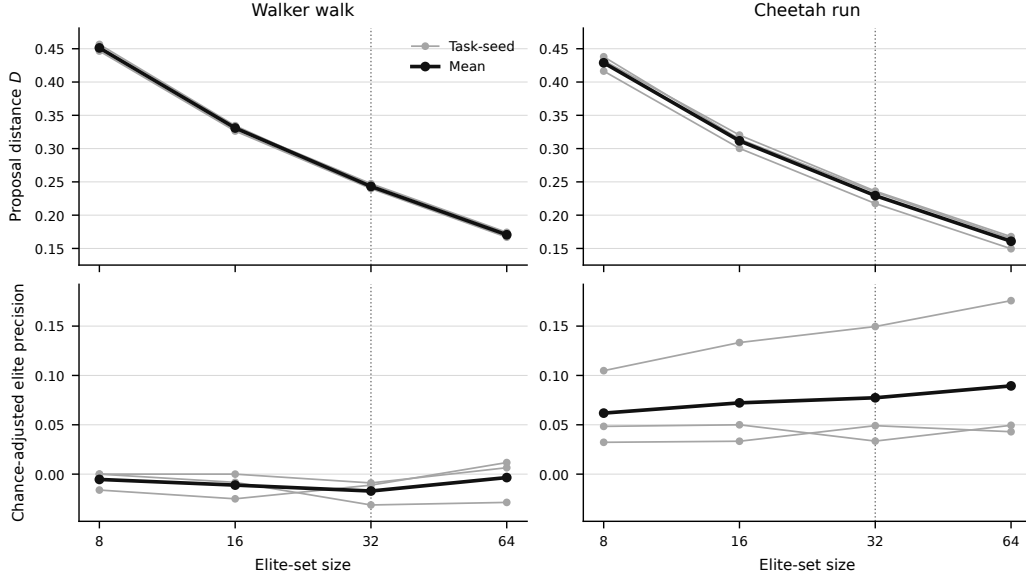


Figure 7: Final-iteration proposal distance and chance-adjusted elite precision for 8, 16, 32, and 64 elites on the original seeds 0–2, recomputed from the same saved pools. Grey lines are task–seed units, black lines are task means, and the dotted line marks the fixed cut of 32. Distance decreases as more candidates enter the refit; task-mean adjusted precision stays below 0.10 at every cut.

G Clipping diagnostics

Candidate actions are clipped to the environment bounds after Gaussian sampling. Table 8 records the raw displacement of the pool mean caused by clipping and the final proportion of proposal coordinates at the width floor. Figure 8 records the fraction of candidates with at least one clipped coordinate along each generator trace for the original seeds 0–2. The displacement and clipping rate measure different effects: many clipped coordinates can produce little mean displacement when their shifts cancel.

Measurement	Walker	Cheetah
Initial mean displacement (action units)	0.116	0.137
Final mean displacement (action units)	0.005	0.006
Final width-floor occupancy (%)	1.2	1.4

Table 8: Bound-related diagnostics for the combined six-seed audit. Clipping displacement compares each pool’s means before and after clipping, in root-mean-square action units. Measurements average model families and states within task–seed units, then seeds within tasks. Mean displacement decreases, and most final proposal coordinates remain above the width floor. The action bounds and floor are fixed across the traces.

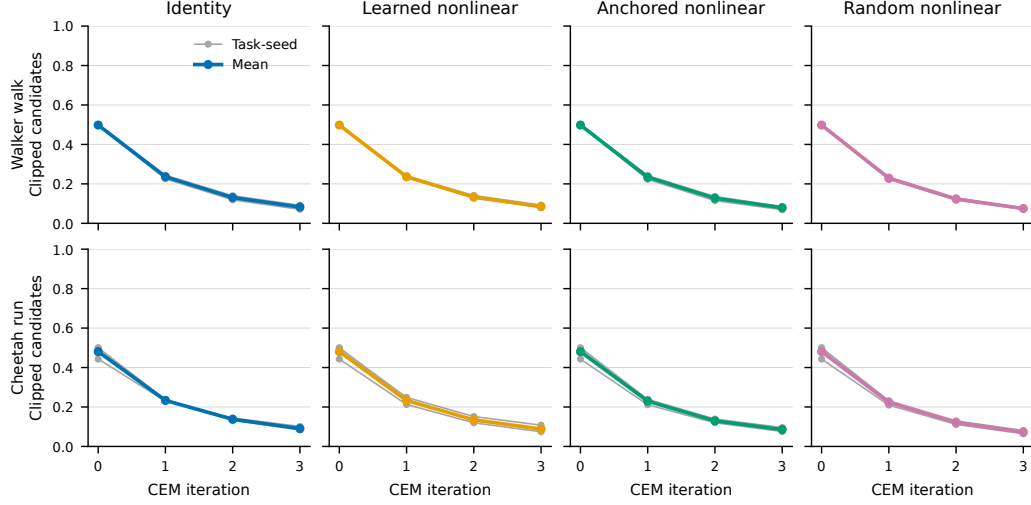


Figure 8: Fraction of candidates with at least one clipped action coordinate, separated by task and pool generator for the original seeds 0–2. CEM settings and base samples are fixed across generators. Every task–generator mean decreases at each update, from 0.48–0.50 at iteration zero to below 0.10 at iteration three.

H Random-elite reference

The magnitude of D depends on the proposal and the finite candidate pool. As a reference, we draw 4,096 random subsets of 32 candidates from each saved pool in the original seeds 0–2, sampling without replacement within each subset. Fitting each subset produces a pool-specific distribution of proposal distances from the environment-elite update.

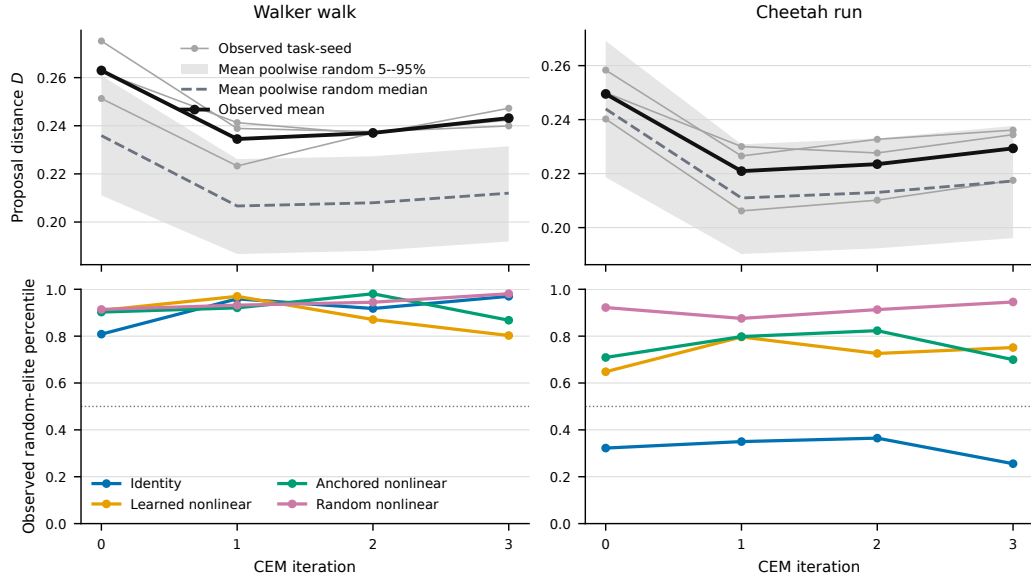


Figure 9: Observed diagonal proposal distances against 32-candidate random refits from the same saved pools for the original seeds 0–2. The upper row compares task–seed values with the mean random median and poolwise 5–95% interval; the lower row gives each surrogate’s random-reference percentile. The observed refit lies above the random median in 28 of 32 task–surrogate–iteration summaries; Cheetah Identity supplies the four exceptions. The reference scales D within a finite pool and does not measure realised cost.

I Reproducibility settings

Item	Setting
Runtime	Python 3.11.3; PyTorch 2.10.0; <code>dm_control</code> 1.0.37; MuJoCo 3.5.0; CUDA 12.8.0.
Dreamer architecture	Encoder channels 32–64–128–256; categorical RSSM with deterministic width 256 and 32×32 stochastic categories; context width 1280; actor and critic hidden width 256.
Dreamer training	500k environment steps per task–seed; five prefill episodes; replay capacity 500k; Adam with learning rates 3×10^{-4} (world model) and 10^{-4} (actor, critic), no weight decay; batches of eight length-16 sequences; one update per environment step; gradient clip 100. KL, reconstruction, reward, and continuation weights are 1; free nats is 1. The imagination horizon is 15, with $\gamma = 0.99$, $\lambda = 0.95$, and actor entropy weight 3×10^{-4} . The final <code>latest.pt</code> checkpoint is frozen.
Surrogate data	32 deterministic frozen-actor episodes per task–seed, shuffled with NumPy seed 0 and split 26/3/3 for training/validation/holdout; length-32 sequences; mean latent inference; rollout targets through horizon 15.
Lift networks	Identity: width 1280. Learned nonlinear: 1280 : 512 : 512 : 256. Anchored nonlinear: affine 1280 : 96 concatenated with 1280 : 480 : 512 : 160. Random nonlinear: fixed 1280 : 512 : 512 : 256. MLP hidden layers use SiLU; outputs are linear.
Shared surrogate layers	Transition $z_{t+1} = Az_t + Ba_t + b$. Compressed-family decoder 256 : 512 : 512 : 1280; Identity uses the identity decoder. Reward and continuation heads are $d_z : 256 : 256 : 1$.
Surrogate optimisation	AdamW; learning rate 3×10^{-4} ; weight decay 10^{-5} ; batch size 128; gradient clip 100; 100 epochs without early stopping. Validation runs each epoch; the retained checkpoint minimises validation decoded-context prior-rollout loss.
Surrogate loss	Horizons $\{1, 3, 5, 8, 15\}$, each weighted 1. Latent-state rollout has weight 1; the summed posterior and prior decoded-context terms have weight 1. Reward MSE and continuation binary cross-entropy have weight 0.25 each. Direct context reconstruction, variance, and covariance terms have weight 0.
Task readout	Linear ridge regression with intercept from decoded context to task variables; coefficient 10^{-3} , intercept unregularised. Four deterministic actor episodes use seeds 120000–120003.
Planning cost	Discount $\gamma = 1$, terminal scale $\lambda_T = 2.5$, action weight $R = 0.01I_6$, and actor-deviation weight 0. Predicted and realised costs use the same weighted squared tracking error.
Elite and pairwise ties	Elite ties use candidate index order. Pairwise agreement excludes realised-cost differences of at most 10^{-12} ; a model tie against a strict realised ordering receives no agreement credit.
Walker weights	$Q = \text{diag}([8, 1^{\times 14}, 2^{\times 9}])$ on height, orientations, and velocity, against the fixed actor-visited target.
Cheetah weights	$Q = \text{diag}([1^{\times 8}, 3^{\times 9}])$ on position and velocity, against the matched frozen-actor reference trajectory.

Table 9: Settings needed to reproduce the frozen models, surrogate fits, readouts, and costs used by the audit. Architecture strings list layer widths from input to output.

Compute required for reproduction. Training the twelve backbones, fitting the 48 surrogates and running the reported audits and splices requires approximately 179 GPU-hours in total. The breakdown is 133 hours for the twelve 500k-step Dreamer backbones, 4.6 hours for data exports and surrogate fits, and 41 hours for the crossed audit and twelve splice units. The first two estimates allocate 12/48 of the corresponding 48-unit training budgets to the units used here. The audit and splice use four CPU cores, 32 GB RAM and one NVIDIA L40S per unit. Walker and Cheetah crossed units take approximately 1.5 and 3.1 hours, respectively. The six held-out splice units account for 18,432 candidate executions and 2.6 of the 41 GPU-hours.

These figures sum wall-clock device time on H200, H100, A100, L40S and V100 GPUs without adjusting for their different speeds. They estimate the cost of reproducing the reported experiment and exclude unused tasks, other evaluations, failed jobs and exploratory runs.

Code and data. The code, trained checkpoints and saved simulator states are not public with this preprint. We plan to release the code and reproduction instructions with an archival publication.